FIXED INCOME RESEARCH | BUY-SIDE ANALYTICS

INVESTMENT RESEARCH REPORT

Fixed Income Risk Decomposition Framework

Separating Rate Risk from Credit Risk in Corporate Bonds

INSTRUMENT	CUSIP	MARKET PRICE
Apple Inc. 4.650% Senior Notes due 2044	037833DV1	$931.78
YTM	**OAS**	**RATING**
5.24%	73 bps	AA+ / Aaa

VALUATION DATE	CLASSIFICATION	ANALYST
2026-02-27	Investment Grade | Technology	Fixed Income Analytics Desk

01 EXECUTIVE SUMMARY

OVERWEIGHT | Risk-Adjusted Return Favors Current Entry

Macro Context

The current environment is characterized by elevated but plateauing interest rates following the most aggressive tightening cycle since the 1980s. The Federal Reserve has maintained the Federal Funds Rate in restrictive territory, with market consensus pricing cuts over the next 12 months. Investment-grade credit spreads remain historically contained.

Investment Problem

Apple Inc. 4.650% Senior Notes due 2044 currently trade at a YTM of 5.24%, implying an OAS of 73bps over the 18-year Treasury. The central analytical question is: **is the bond's current discount to par principally a function of higher risk-free rates, or does it embed meaningful credit risk premium?**

Key Findings

Finding	Summary
Rate Dominance	Modified duration of 11.85x implies that ~10.9% of bond value is at risk per 100bps rate shock.
Credit Benign	Current OAS of 73bps sits with a Z-score of -1.19σ relative to the 12-month average.
Asymmetric Upside	Under a rate-cut scenario, expected total return is +20.6% vs. -19.6% under adverse stagflation.
Carry Attractive	With 73bps of carry over Treasuries, the breakeven spread widening before carry is consumed is ~1185b

Investment Implication

Given the rate-dominated risk profile, the bond is best suited for portfolios with constructive views on rate normalization. Duration extension at current levels offers asymmetric payoff.

02 THEORETICAL FRAMEWORK

2.1 Bond Price Structure

A fixed-rate bond's price is the present value of its contractual cash flows, discounted at the market yield-to-maturity (YTM).

$$P = SUM [C/2 / (1 + y/2)^{t}] + F / (1 + y/2)^{N}$$

2.2 Yield Decomposition

The yield-to-maturity of a corporate bond can be decomposed into two distinct components:

YTM = Risk-Free Rate (Treasury) + Credit Spread (OAS)

This decomposition is fundamental to risk attribution. Rate risk dominates high-grade corporate bonds.

2.3 Duration and Convexity

Metric	Formula (Conceptual)	Interpretation
Macaulay Duration	Weighted avg. time to cash flows	Time-weighted payback period
Modified Duration	MacD / (1 + y/m)	%Price change per 100bps yield change
Dollar Duration (DV01)	ModD × Price / 10,000	Dollar price change per 1bp
Convexity	2nd derivative of price/yield	Non-linearity; positive = favorable curvature

03 INSTRUMENT DESCRIPTION

3.1 Security Specifications

Parameter	Detail	Parameter	Detail
Issuer	Apple Inc.	Coupon	4.650% Fixed
CUSIP	037833DV1	Frequency	Semi-Annual
Settlement	2026-02-27	Day Count	ACT/ACT
Maturity	February 23, 2044	Years to Maturity	17.98 yrs
Remaining Coupon Periods	36	Face Value	$1,000
Rating (S&P)	AA+	Rating (Moody's)	Aaa
Sector	Technology	Currency	USD

3.2 Current Market Data

Market Price (Clean)	YTM	10Y Treasury	OAS	DV01	Modified Duration
$931.78	5.2400%	4.51%	73 bps	$1.103	11.8470x

3.3 Rationale for Instrument Selection

Apple Inc. represents the quintessential investment-grade benchmark issuer: AA+ rated with fortress balance sheet, near-zero default probability over the investment horizon.

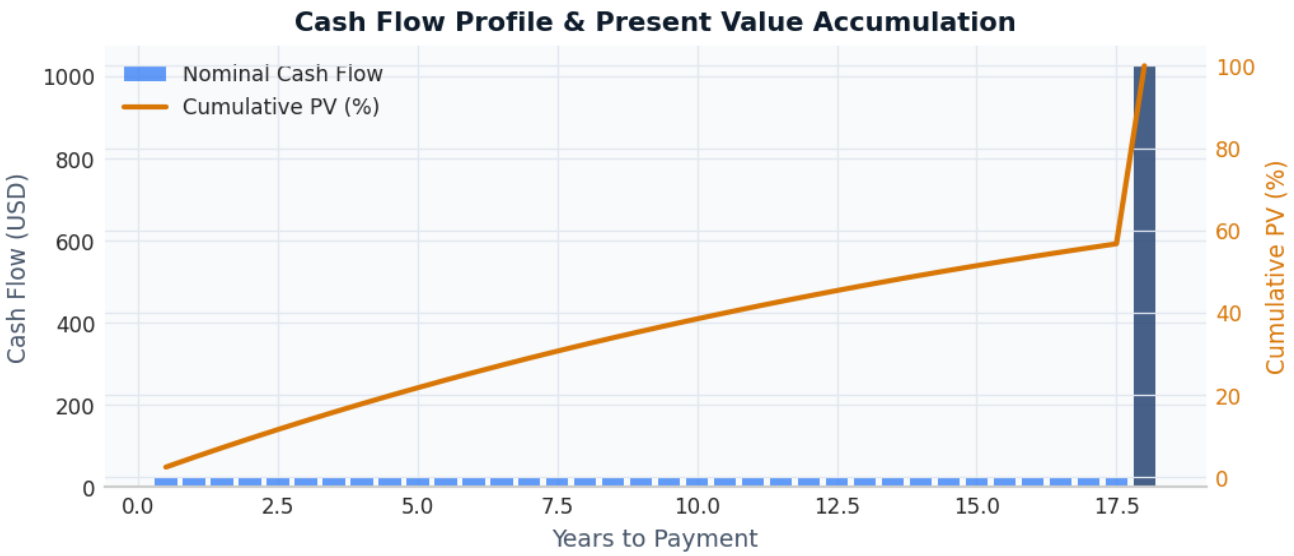


Figure 1: Cash flow profile and cumulative PV accumulation over 36 semi-annual periods.

04 BASE VALUATION

4.1 Cash Flow Construction & Pricing

Period	Time (Yrs)	Cash Flow ($)	Discount Factor	PV ($)
1	0.50	$23.25	0.974469	$22.66
2	1.00	$23.25	0.949590	$22.08
3	1.50	$23.25	0.925346	$21.51
4	2.00	$23.25	0.901721	$20.97
5	2.50	$23.25	0.878699	$20.43
6	3.00	$23.25	0.856264	$19.91
...	...	...	...	...
36	18.00	$1,023.25	0.394137	$403.30
TOTAL				$931.7826

4.2 Pricing Summary

Clean Price	Accrued Interest	Dirty Price	YTM (Computed)	YTM vs. Market	Validation
$931.7826	$0.5138	$932.2965	5.2400%	+0.0 bps	✓ PASS

The model price of $931.78 is derived by discounting all semi-annual cash flows at the market YTM of 5.24%.

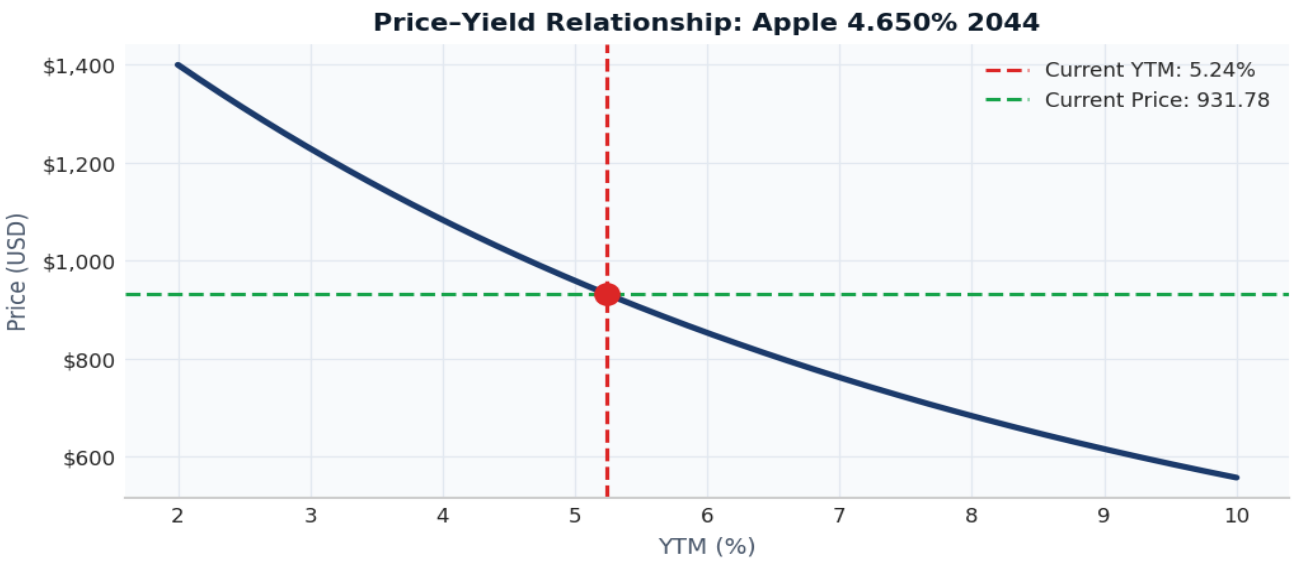


Figure 2: Full price-yield curve showing the convex relationship.

05 INTEREST RATE SENSITIVITY ANALYSIS

5.1 Key Risk Metrics

Macaulay Duration	Modified Duration	Convexity	DV01
12.157 yrs	11.847x	184.049	$1.103
Weighted avg. payback	%Δ Price / 100bps	2nd order adjustment	Per $1,000 face

5.2 Parallel Shock Analysis

Shock (bps)	New YTM (%)	New Price ($)	Exact ΔP (%)	Duration Approx (%)	Conv. Adj. (%)	Error (bps)
-200	3.240%	$1,191.17	+27.837%	+26.913%	+27.375%	-46.2
-100	4.240%	$1,051.26	+12.822%	+12.712%	+12.767%	-5.5
-50	4.740%	$989.18	+6.160%	+6.147%	+6.154%	-0.7
-25	4.990%	$959.92	+3.020%	+3.018%	+3.019%	-0.1
+25	5.490%	$904.71	-2.905%	-2.903%	-2.904%	+0.1
+50	5.740%	$878.67	-5.700%	-5.687%	-5.693%	+0.6
+100	6.240%	$829.50	-10.977%	-10.876%	-10.927%	+5.0
+200	7.240%	$741.71	-20.399%	-19.627%	-20.013%	+38.6

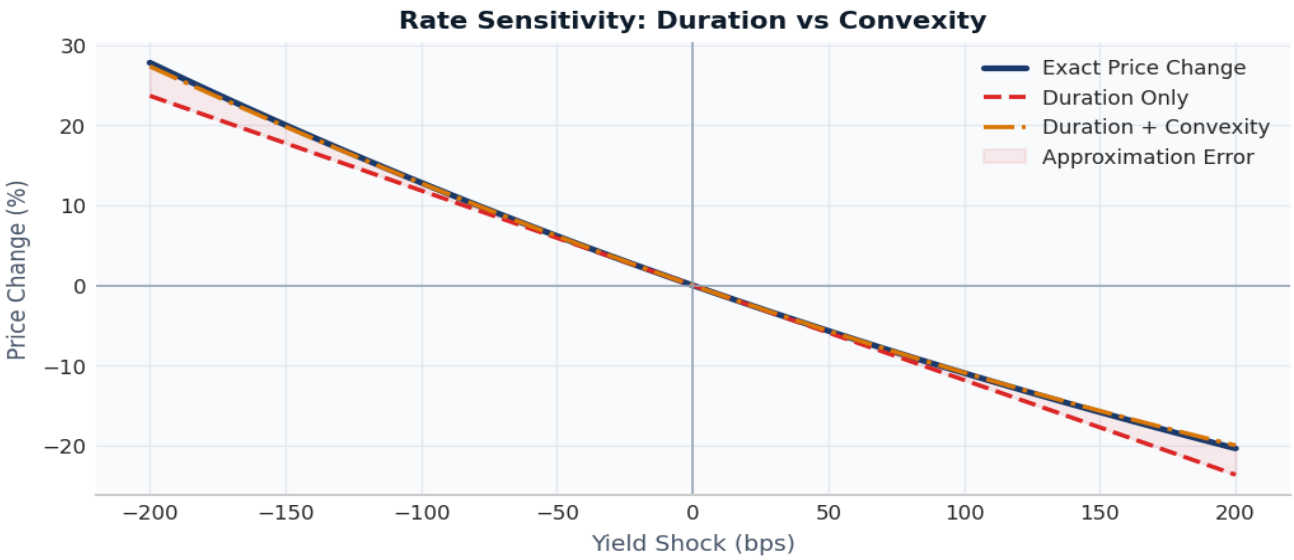


Figure 3: Rate sensitivity comparison. The gap between Duration-Only and Exact curves demonstrates convexity benefit.

06 CREDIT SPREAD SENSITIVITY ANALYSIS

6.1 OAS Shock Analysis

Spread Shock (bps)	New OAS (bps)	New YTM (%)	New Price ($)	Price Change (%)
-50	23	4.740%	$989.18	+6.160%
-25	48	4.990%	$959.92	+3.020%
+25	98	5.490%	$904.71	-2.905%
+50	123	5.740%	$878.67	-5.700%
+75	148	5.990%	$853.61	-8.389%
+100	173	6.240%	$829.50	-10.977%
+150	223	6.740%	$783.94	-15.866%
+200	273	7.240%	$741.71	-20.399%

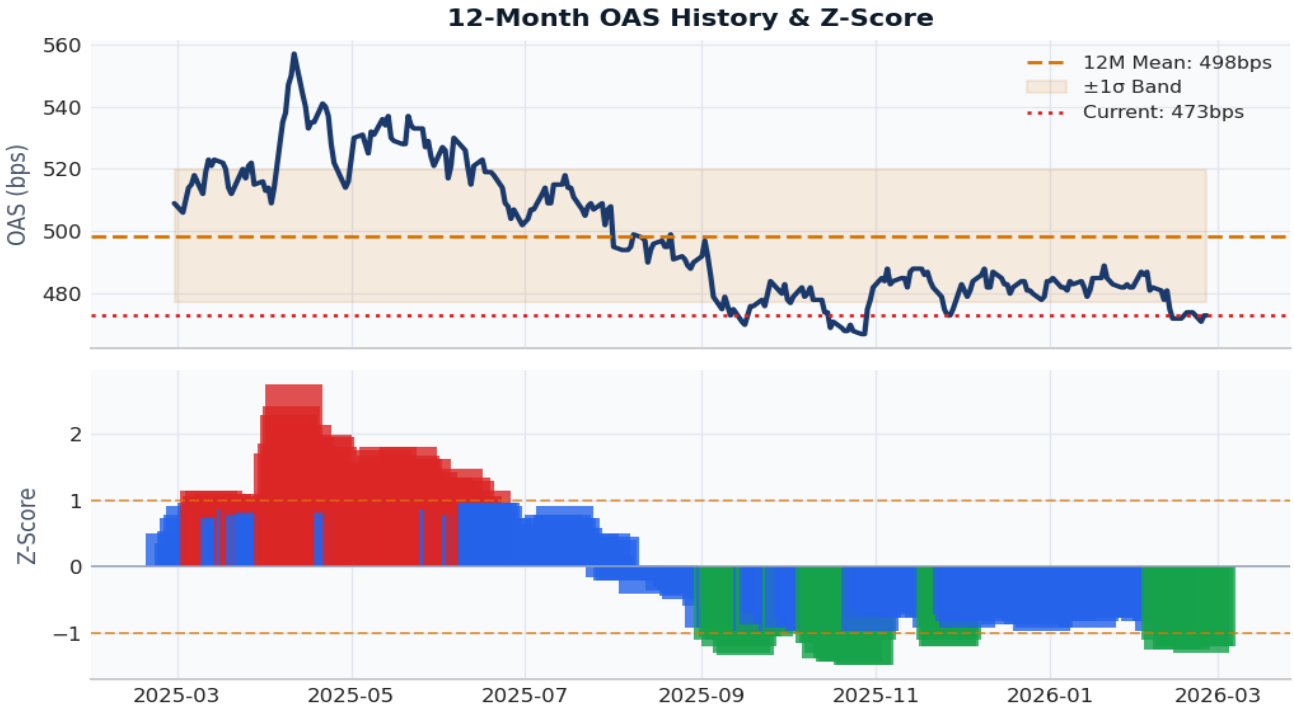


Figure 4: 12-month OAS history and Z-score. Current spread of 473bps shows a Z-score of -1.19σ.

07 RETURN DECOMPOSITION — TRAILING 12 MONTHS

7.1 Yield Attribution

Component	12M Ago	Current	Change (bps)	Price Attribution
10Y Treasury Yield	4.240%	4.050%	-0.2 bps	+1.93%
OAS / Credit Spread	509.0 bps	473.0 bps	-36.0 bps	+54.52%
YTM (Total)	9.330%	8.780%	-0.5 bps	+56.46%
Coupon Income	—	—	—	+7.81%
TOTAL RETURN				+64.26%

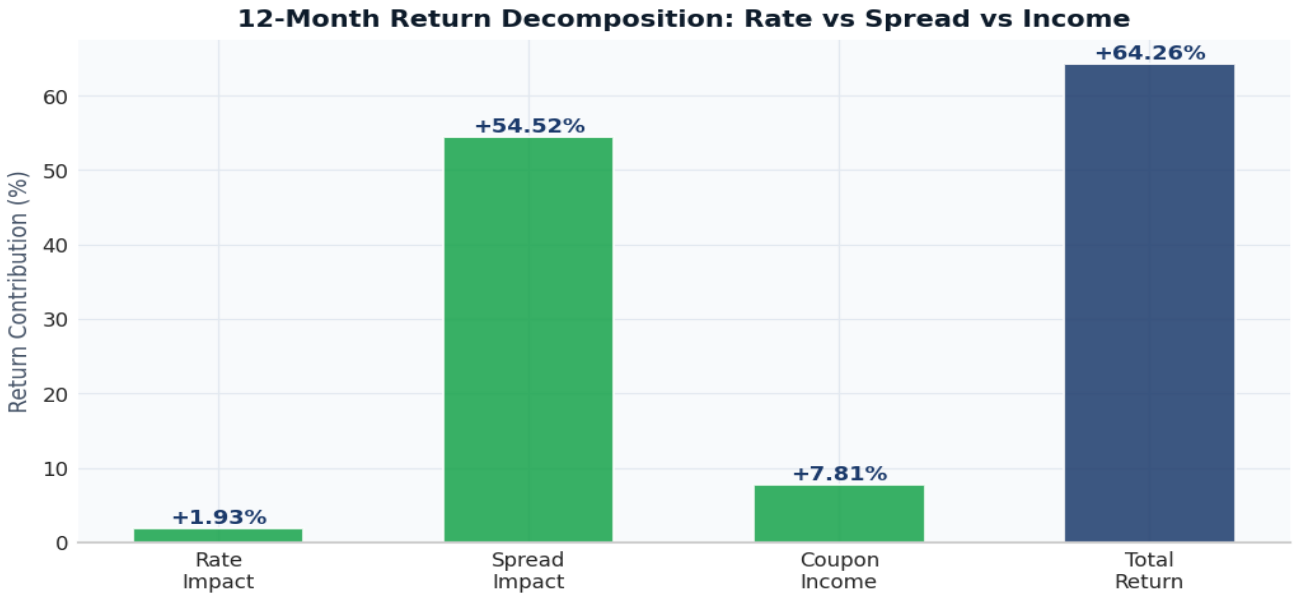


Figure 5: Waterfall decomposition of 12-month return into rate, spread, and income components.

08 YIELD CURVE CONTEXT

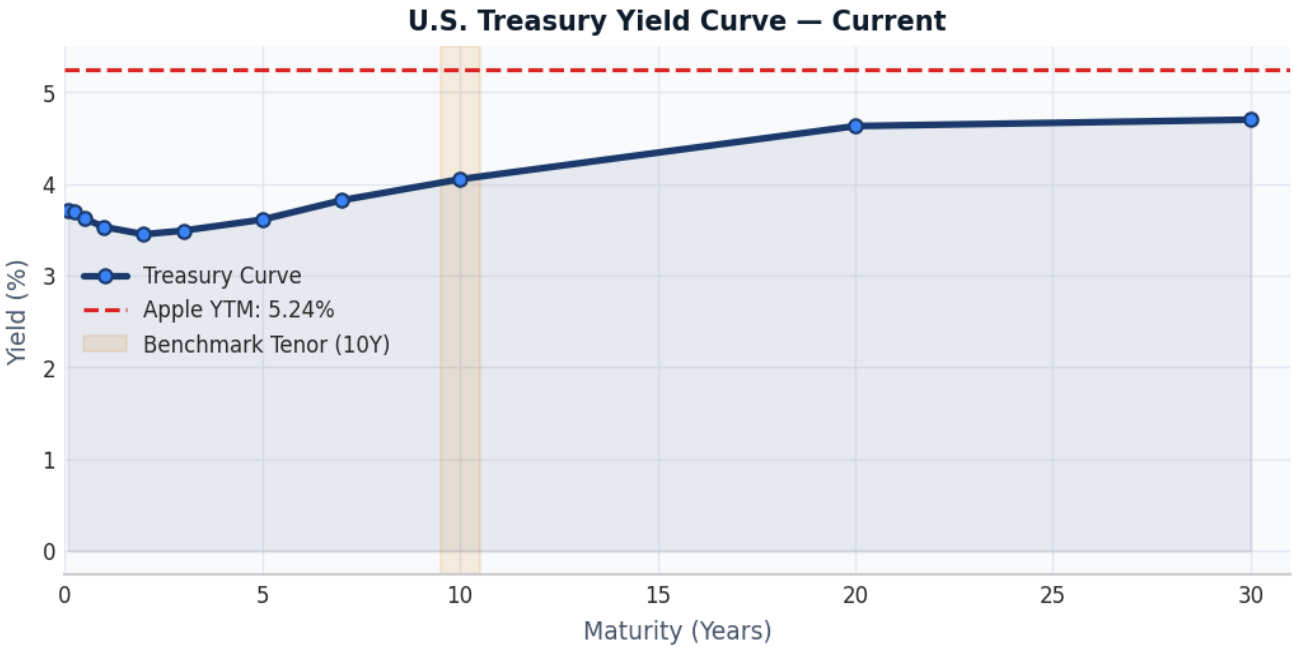


Figure 6: Current U.S. Treasury yield curve. The Apple YTM benchmark is plotted for reference.

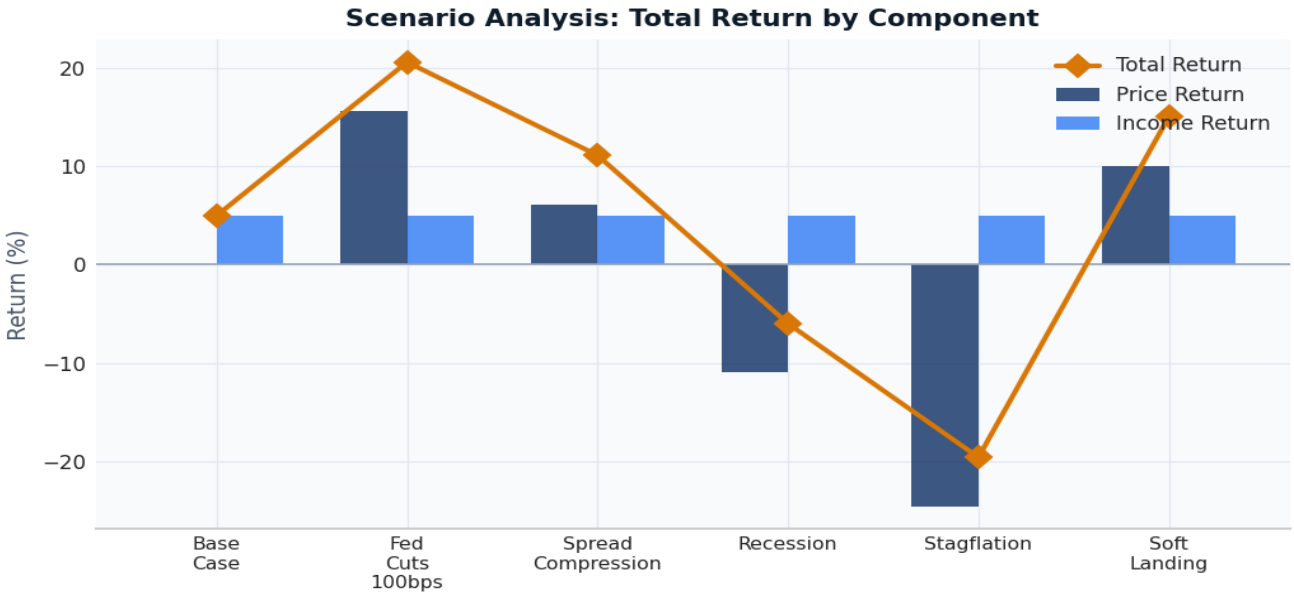
Treasury Curve Data

Tenor	Yield (%)	Tenor	Yield (%)	Tenor	Yield (%)
1M	3.710%	3M	3.690%	6M	3.620%
1Y	3.530%	2Y	3.450%	3Y	3.490%
5Y	3.610%	7Y	3.820%	10Y	4.050%
20Y	4.630%	30Y	4.700%	—	—

09 SCENARIO ANALYSIS

Scenario Return Summary

Scenario	New YTM (%)	New Price ($)	Price Return	Income Return	Total Return	Assessment
Base Case	5.240%	$931.78	+0.00%	+4.99%	+4.99%	Hold
Fed Cuts 100bps	4.040%	$1,077.49	+15.64%	+4.99%	+20.63%	Attractive
Spread Compression	4.740%	$989.18	+6.16%	+4.99%	+11.15%	Attractive
Recession	6.240%	$829.50	-10.98%	+4.99%	-5.99%	Cautious
Stagflation	7.740%	$702.53	-24.60%	+4.99%	-19.61%	Avoid
Soft Landing	4.440%	$1,025.84	+10.09%	+4.99%	+15.08%	Attractive



10 RISK-ADJUSTED METRICS

10.1 Return and Risk Summary

Metric	Value	Benchmark / Context
Expected Return (Soft Landing)	+15.08%	Base-bull scenario, 12-month horizon
Best Case Return (Rate Cut)	+20.63%	Fed -100bps scenario
Downside Return (Stagflation)	-19.61%	Tail risk scenario
Upside / Downside Ratio	1.05x	Bull return / Bear loss magnitude
Current OAS	73 bps	12M avg: 498bps
OAS Z-Score	-1.19σ	Negative = tighter than avg
Annual Coupon Carry	4.650%	Gross income return
Carry over Treasury (bps)	72.7 bps	OAS proxy
Breakeven Spread Widening	1184.7 bps	Before carry is consumed

11 INVESTMENT CONCLUSION

OVERWEIGHT

12-Month Target Return: **8–10%**
Under base-to-bull scenarios (Soft Landing / Rate Cuts)
Risk: Material only under stagflation tail scenario

Positioning Recommendation

Dimension	Recommendation	Rationale
Position Size	Overweight vs Benchmark	Favorable risk/reward at current OAS/duration
Duration	Maintain / Extend	Constructive on rate normalization
Hedge	No Rate Hedge Recommended	Duration exposure is the thesis driver
Stop Loss	10Y Treasury > 5.00%	Reassess if rates break above 5%
Take Profit	10Y Treasury < 3.75%	Reduce on large rally; re-evaluate entry
Review Trigger	OAS > 150 bps	Credit signal; investigate fundamentals

12 MODEL LIMITATIONS

Parallel Yield Shifts Only	The scenario analysis assumes parallel shifts in the yield curve.
No Liquidity Premium	The model does not explicitly decompose the OAS into default risk premium and liquidity premium components.
No Explicit Default Modeling	This framework does not model explicit default scenarios.
Static Duration Assumption	Duration is calculated at current YTM and assumed constant across scenarios.

13 TECHNICAL APPENDIX

A.2 Model Parameters Summary

Parameter	Value	Parameter	Value
Coupon Rate	4.650%	Face Value	$1,000
Settlement Date	2026-02-27	Maturity Date	2044-02-23
Years to Maturity	17.9849	Periods	36
Market YTM	5.2400%	Treasury Yield	4.513%
OAS	73 bps	Model Price	$931.7826
Macaulay Duration	12.157378 yrs	Modified Duration	11.846987
Convexity	184.049090	DV01	$1.103024

DISCLAIMER: This report is produced by an analytical model for educational and research purposes.